



control units



mindy TT1L

Instructions and warnings for the fitter

Istruzioni ed avvertenze per l'installatore

Instructions et recommandations pour l'installateur

Anweisungen und Hinweise für den Installateur

Instrucciones y advertencias para el instalador

Instrukcje i uwagi dla instalatora

Instructies en waarschuwingen voor de installatie



COMPANY
WITH QUALITY SYSTEM
CERTIFIED BY DNV
=ISO 9001/2000=

Nice

Warnings

The TT1L control unit is suitable for the control of a single-phase electrical device powered by the electrical mains, for example, lamps for illumination, motors for irrigation etc.

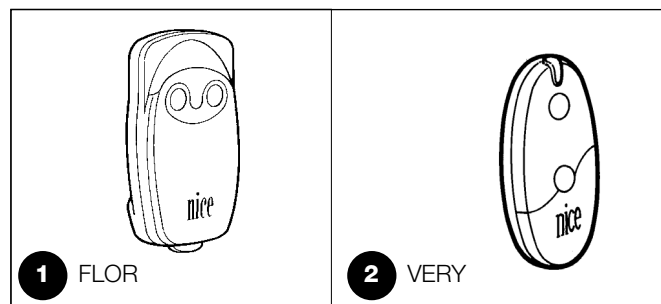
Any other use is improper and prohibited. The unit must be installed by qualified technicians in compliance with the electrical and safety regulations currently in force.

1) Product description

The TT1L control unit enables the user to switch electrical appliances on and off using a remote control device. The max. power of this device must be no greater than 500W (400VA), and powered by electrical mains voltage. Possible commands are ON/OFF, Man Present and Timed Switch Off (with 2 independent timers).

The control unit incorporates a radio receiver operating at a frequency of 433.92 MHz, with rolling code technology that guarantees high levels of security. Up to 30 transmitters ("FLOR" fig. 1, "VERY" fig. 2) can be memorized for each control unit. Two keys are used on each transmitter, one for the Switch ON command and the other for the Switch OFF command. Any kind of programming can be carried out directly using the transmitters. A beeping noise will guide the user through the various phases.

N.B.: The control unit can manage other kinds of transmitters and functioning modes, too. Please refer to Chapter 4 "Additional information" for further explanations.



2) Installation

⚠ The electrical systems and automations must be installed by qualified and experienced personnel in compliance with current legislation. Before you proceed to make any connections make sure that the power supply is disconnected.

1. Strip the cable of the electric device to be controlled and the power supply cable about 3 cm and then the single wires approx. 5 mm.
2. Open the container by removing the "cable cap" as shown in fig. 4.
3. Thread the two cables through the designated holes in the "cable cap" (see fig. 5).
4. Pull the card a few centimetres out of the container (see fig. 6).

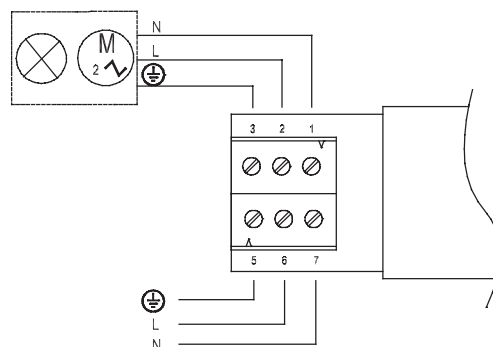
5. Connect the wires to the terminals as shown in fig. 7, observing the diagram in fig. 3 and the operations described in chapter 2.1.
6. Fold the cables as shown in fig. 9. 8.
7. Push the card inside the container, make sure that the stripped length of the cable is fully inside the container, then slide the "cap" on the cables until the container closes completely (see fig. 9).
8. Fasten the control unit to a suitable surface, protected against impact. Double-sided tape can be used for this. To avoid the risk of water leaks the control unit should be positioned with the cables towards the bottom as shown in Fig. 10. Do not place it with the cables towards the top (Fig. 11).

⚠ Do not perforate the container.

2.1) Electrical connections

⚠ Carefully follow all the connection instructions, if you have any doubts do NOT make experiments but consult the relevant technical specifications sheets which are also available on the web site www.niceforyou.com

An incorrect connection may cause serious damage to the control unit.



2.1.1) Connecting up the electrical appliance

The power supply of the electrical appliance the user wishes to control (max. 500W/400VA) must be connected to output terminals 1-2; no. 3 is the earth terminal.

2.1.2) Power supply

The main supply to the unit must be connected using the terminals 5-6-7 (earth, phase, neutral) as shown in Fig. 3 "Electrical Connections"

⚠ In the control unit, terminal no. 7 (Neutral) is directly connected to output terminal 1. This means that even if it is switched off, the electrical appliance which is to be controlled is not completely disconnected from the electricity power supply line.

3) Programming

Each transmitter is recognised by the control unit by means of an unequivocal "code". A "memorisation" phase must therefore be performed in order to allow the control unit to recognise each single transmitter.

- **All the memorisation sequences are timed, that is, they must be completed within the programmed time limits.**

- **Programming via radio may be done on all the control units within the range of the transmitter; therefore, only the one involved in the operation should be kept switched on.**

Two keys are used on each transmitter: key 1 = "ON" and key 2 = "OFF". In those transmitters with 4 keys, those which remain "empty" can be used to control another control unit: key 3 = "ON" and key 4 = "OFF".

When the memory contains no codes the first radio control unit can be entered as follows:

Table "A1"	Memorizing the first transmitter	Example
1.	As soon as the control unit is powered, 2 long beeps will sound	 
2.	Within 5 seconds press and hold down either key 1 or 2 (or key 3 or 4) on the transmitter to be memorised (for approx. 3 seconds)	 3s
3.	Release the key when you hear the first of the 3 beeps confirming memorisation	

N.B.: If there are already transmitters memorised in the control unit, short beeps will sound when it is switched on. This means that the above procedure is not valid and another memorisation procedure must be used (Table "A2").

When one or more transmitters have already been memorised, others may be enabled as follows:

Table "A2"	Memorizing other transmitters	Example
1.	Press and hold down either key 1 or 2 (or key 3 or 4) on the new transmitter until you hear a beep (after about 5 seconds), then release it	New   5s
2.	Press any key of a just functioning transmitter slowly 3 times	Old  X3
3.	Press either key 1 or 2 (or key 3 or 4) on the new transmitter again and release it when you hear the first of the 3 beeps.	New  

N.B.: If the memorization procedure is successful 3 long beeps will sound. 6 beeps will sound when the memory is full (30 transmitters), telling you the transmitter cannot be memorised.

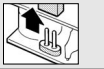
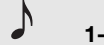
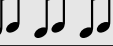
If you need to delete all the data contained in the memory of the control unit, carry out the procedure on table "A3".

The memory can be deleted:

- with a **non-memorised** transmitter by beginning the procedure from point A, even if a transmitter which has already been memorised is necessary at point 1.
- using a **previously memorised** transmitter starting the procedure from point N. 1

The following can be erased:

- only the transmitter codes, finishing at point N. 4
- All the data (transmitters and timer programming), completing the procedure up to point N. 5.

Table "A3"	Erasing the memory	Example
➔ A	Switch the control unit off and cut the jumper on the weld side of the card (see fig. 12). The jumper must be reconstituted when the deletion procedure has been completed.	 
B	Power the control unit and wait for the initial beeps	  1-3
➔ 1	Press and hold down key 2 (or 4) of a previously memorized transmitter until you hear a beep (after about 5 seconds) then release it.	 5s
2	After 1 second press the same key again and release it exactly during the third beep.	 
3	After 1 second press the same key again and release it exactly during the third beep	 
➔ 4	After 1 second press the same key again and release it exactly during the third beep.	 
5	If you wish to delete all the data in the memory, press the same key again within 2 seconds until you hear the first of 5 beeps, then release them	 

N.B.: 5 beeps will indicate that all the codes in the memory have been deleted.

4) Additional Information

Besides the transmitters from the “FLOR” and “VERY” series, the control unit will also recognise other kinds of transmitters manufactured by Nice (please refer to Chapter 4.1 “Usable transmitters”). Moreover, by using a

special memorisation procedure, it is possible to associate a special command to each of the transmitter keys (please refer to Chapter 4.2 “Programming transmitters in Mode I and Mode II”).

4.1) Usable transmitters

The transmitters which can be used are shown in Table “A4”, together with their relative radio coding.

Table “A4”		
CODING		Transmitters
FLOR	Rolling code	FLO1R - FLO2R - FLO4R
		VERY VR
		ERGO1 - ERGO4 - ERGO6
		PLANO1 - PLANO4 - PLANO6 - PLANO TIME
SMILO	Rolling code	SM2 - SM4
FLO	Fixed cod	FLO1 - FLO2 - FLO4
		VERY VE

The transmitters have different kinds of coding, and the control unit is unable to recognise them simultaneously. The first transmitter memorised will thus determine the type of coding, and consequently the transmitters which can be memorised successively.
All codes must be deleted if the user wishes to change the type of transmitter (see tables “A3”).
The type of code can be checked by counting the number of beeps which sound when the control unit is switched on.

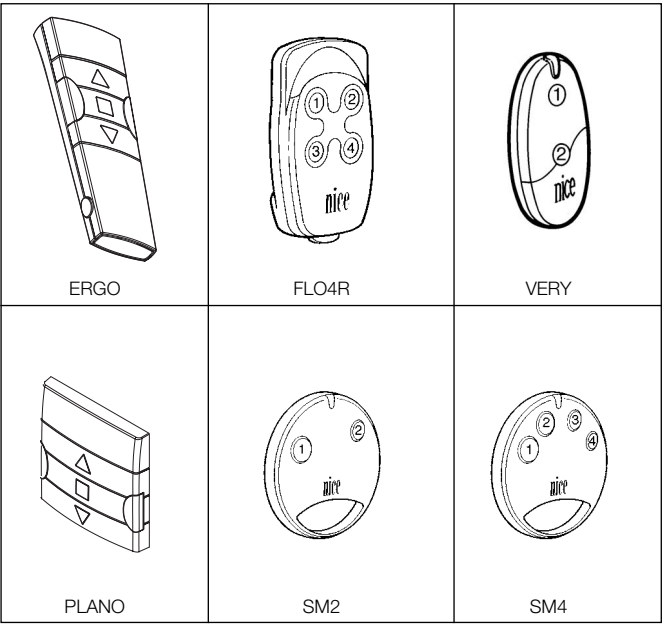


Table “A5”		
BEEP		Types of coding for the transmitters memorised
1 short beep	♪	Transmitters with FLO coding
2 short beeps	♪ ♪	Transmitters with FLOR coding
3 short beeps	♪ ♪ ♪	Transmitters with SMILO coding
2 long beeps	♪♪	Empty memory (no memorized transmitter)

4.2) Memorizing the transmitters in Mode I and Mode II

Tables “A1” and “A2” show the memorisation of transmitters in Mode I
The transmitters can also be memorised in Mode II. This mode gives the user the most flexibility when using the transmitters and the control unit.

Transmitters in both Mode I and Mode II can be memorised in the same control unit.
The differences between the 2 programming modes have been described below:

4.2.1) Mode I

In this mode, the command associated with the transmitter keys is fixed: key 1 commands the switching ON, key 2 commands the switching OFF. If keys 3 and 4 are on the same transmitter, they can be used to command another control unit: key 3 controls Switching On = (ON) and key 4 switching off = (OFF).
The user will carry out a single memorisation phase for each transmitter on the control unit. **It is not important whether keys 1 or 2 are pressed (or keys 3 or 4) during this phase.**

Please refer to Chapter 3 “Programming” for information about how to memorise or delete the transmitters.

Example	memorization mode I	
Key 1	ON TT1L no.1	
Key 2	OFF TT1L no.1	
Key 3		ON TT1L no.2
Key 4		OFF TT1L no.2

4.2.2) Mode II

This mode enables the user to associate one of the following commands to each key of the transmitter:

- 1 "ON/OFF" (the same key switches on and then off, etc.)
- 2 "MAN PRESENT" (the appliance will only function as long as the key is pressed down)
- 3 "TIMER1" after being switched on, the appliance will switch off at the time set (factory setting: 1 minute)
- 4 "TIMER2" after being switched on, the appliance will switch off at the time set (factory setting: 10 minutes).

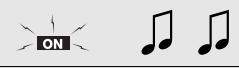
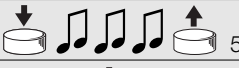
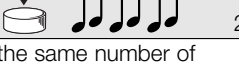
A memorisation phase for each of the keys on the transmitter must be carried out on the control unit. **During this phase, the user must press the key the command** is to be associated with. If the user wishes to associate another command to another key on the transmitter, s/he must carry out another memorisation phase.

The keys on the same transmitter can be associated either with more commands of the same control unit or with commands from different control units.

Example 1	memorization mode II	
Key 1	ON/OFF	on TT1L no.1
Key 2	TIMER 1	on TT1L no.1
Key 3	TIMER2	on TT1L no.1
Key 4	MAN PRESENT on TT1L no. 2	

Example 2	memorization mode II		
Key 1	ON/OFF	on TT1L no.1	
Key 2		ON/OFF	on TT1L no.2
Key 3			ON/OFF on TT1L no.3
Key 4			TIMER1 on TT1L no.3

When the memory contains no transmitters the first one can be memorized in mode II as follows:

Table "A6"	Memorizing the first transmitter in mode II	Example
1.	As soon as the control unit is powered, 2 long beeps will sound	
2.	Press the transmitter key to be memorised within 5 seconds, and hold it down until you hear all 3 beeps , then release.	
3.	Within 3 seconds press the same key of the transmitter as many times as the number corresponding to the desired command: 1 = "ON/OFF" 2 = "MAN PRESENT" 3 = "TIMER1" 4 = "TIMER2"	
4.	After approximately 3 seconds, the user will hear a certain number of beeps. These beeps correspond to the command selected.	
5.	Press the same key again within 2 seconds to confirm programming. Release this key when you hear the first of the 3 beeps.	

N.B.: If the memorization procedure is successful 3 long beeps will sound. If, at point 4, the user does not hear the same number of beeps as the command required, wait for a few seconds to exit the procedure, without confirming the memorisation.

When one or more transmitters have already been memorised, others may be enabled in Mode II as follows:

Table "A7"	Memorizing other transmitters in Mode II	Example
1.	Press and hold down the new key to be memorized until you hear a beep (after about 5 seconds) then release it	New 
2.	Within 5 seconds press and hold down the key of a previously memorized (old) transmitter approx. 5 seconds until you hear 2 short beeps then release it.	Old 
3.	Within 5 seconds press the same key of the (old) transmitter as many times as the number corresponding to the desired command: 1 = "ON/OFF" 2 = "MAN PRESENT" 3 = "TIMER1" 4 = "TIMER2"	Old 
4.	After approximately 3 seconds, the user will hear a certain number of beeps. These beeps correspond to the command selected	
5.	Press the new transmitter key to be memorised again within 2 seconds to confirm programming. Release this key when you hear the first of the 3 beeps.	New 

N.B.: If the memorization procedure is successful 3 long beeps will sound. 6 beeps will sound when the memory is full (30 transmitters), telling you the transmitter cannot be memorised.

A new transmitter can easily be memorised, keeping the same features as the old transmitter, by following the procedure detailed in Table "A9". Any new transmitters which are memorised in this manner will take on the same characteristics as the old one. If the old one was memorised in Mode I, the new one will also operate in Mode I. If, on the other hand, the old one was memorised in Mode II, the key of the new transmitter will also be associated to the same command as the old one.

Table "A8"	Memorizing other transmitters	Example
1.	Hold down the key of the new transmitter you wish to memorise for at least 3 seconds, then release	New 
2.	Press the key of the (old) transmitter you wish to memorise for at least 3 seconds, then release.	Old 
3.	Press the key of the new transmitter again for at least another 3 seconds, then release.	
4.	Press the key of the old transmitter again until 3 beeps sound to confirm the new transmitter has been memorised.	Old 

N.B.: 6 beeps will sound when the memory is full (30 transmitters), telling you the transmitter cannot be memorised.

If you need to delete all the data contained in the memory of the control unit by using a transmitter **memorized in Mode II**, carry out the procedure shown in table "A9".

⚠ Deletion with transmitters memorised in Mode II is a particularly complex procedure and the points below must be observed. The procedure detailed in Table A3 is much simpler if a transmitter which has not been memorised is used.

- Deletion is not possible with transmitters which have a "MAN PRESENT" command. If necessary, a new transmitter and the procedure detailed in Table "A3" must be used. .
- Se si esegue la procedura con trasmettitori con comando TIMER è necessario partire dal punto A, altrimenti iniziare dal punto 1

You can delete:

- the transmitters only, by stopping at point 4 of the procedure;
- all the data (transmitters, TIMER 1 and TIMER2 duration), by carrying out the procedure through to point 5.

Table "A9"	Memory deletion with transmitter memorized in Mode II	Example
➔ A	A Switch the control unit off and cut the jumper on the weld side of the card (see fig. 12). The jumper must be reconstituted when the deletion procedure has been completed.	 
B	Power the control unit and wait for the initial beeps	  1-3
C	Press and hold down the key with the TIMER command. The relay will switch on, and after approximately 3 seconds it will switch off. Proceed to point 2 keeping the key held down.	  3s 
➔ 1	If the relay is OFF, press and release the key with the ON/OFF command in order to switch it on. Press the key again and keep it pressed down.	  5s
2	After approximately 5 seconds you hear a beep, then release	 
3	After 1 second press down the same key again and release during the third beep.	   
4	After 1 second press down the same key again and release during the third beep.	   
➔ 5	After 1 second press down the same key again and release during the third beep.	   
6	If you wish to delete all the data in the memory, press the same key again within 2 seconds until you hear the first of 5 beeps, then release it.	    

N.B.: 5 beeps will indicate that all the codes in the memory have been deleted.

4.3) Auxiliary Programming

The duration of TIMERS 1 and 2 can be programmed on the control unit, as can the use of 3 key transmitters such as "▲ ■ ▼", Please refer to the following chapters.

4.3.1) Programming the timers

The control unit enables the user to programme 2 independent timers for switching the relay off automatically after a preset time: "TIMER1" and "TIMER2". The time is reset whenever the command is used, and it is possible to anticipate Switching Off either by keeping the TIMER command active for more than 3 seconds, or by sending an OFF command.

The factory time for the 2 timers, and the time set after the memory has been deleted completely, are 1 minute (TIMER1) and 10 minutes (TIMER2).

The Switch On time can be programmed from a minimum of 1 second to a maximum of 9 hours.

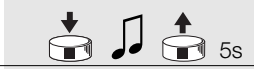
Table "A10"	Programming the duration of the timers	Example
1.	Press and hold down the key on the transmitter relative to the timer you wish to programme. The relay will switch on (ON).	 
2.	Keep the same key held down. The relay will switch off (OFF) after approximately 3 seconds.	  3s
3.	Keep the same key held down until the relay switches on again (this will happen after approximately 8 seconds). The time counting phase will start immediately. Release the key.	   8s
4.	When the time the user wishes to programme has elapsed, s/he must press one of the keys on one of the transmitters which has been memorised in the control unit. 3 long beeps will sound to indicate the time has been memorised, followed by 1 or 2 short beeps to indicate whether programming refers to TIMER1 or TIMER2.	    1-2

Nota: Se si vuole modificare il tempo dei temporizzatori basta ripetere la procedura partendo dal punto 1

4.3.2) Using 3 key type transmitters “▲■▼”

If transmitters have keys like “▲■▼” memorised in Mode I, the “OFF” command can be associated to key ▼. for each one of them. This makes using the “events” forecasted by the transmitters like, for example PLANO TIME, easy. Key ■ will continue carrying out the “OFF” function.

The transmitter must be memorised in Mode I in order to enable this function, and the said function will only be enabled to that transmitter. Repeat programming for all the necessary transmitters.

Table “A11”	FUNCTION “ON” “OFF” with ▲■▼	Example
1.	Press and hold down key ■ of a previously transmitter memorised in Mode I until you hear a beep (after about 5 seconds) then release it.	 5s
2.	Within 3 seconds press simultaneously the two keys ▲ ▼ and hold them down for approx. 5 seconds until 3 beeps sound to confirm the function has been programmed.	 3s

N.B.: If the user should wish to exclude the function, s/he must repeat the procedure starting from point 1.

5) What to do if... a short troubleshooting guide!!

When the control unit is switched on, no beep sounds and the transmitters do not transmit any commands.

Make sure that the control unit is powered correctly: there must be mains voltage between terminals 6-7. If the power supply is correct, there is likely to be a serious fault in the system, and the control unit will have to be replaced.

6 beeps will sound after a radio-controlled command. The relays will neither switch on nor off.

The radio control is out of synch, the transmitter memorization procedure must be repeated.

After a radio command, you hear 10 beeps and then the relais switches on

The self diagnosis of the parameters in the memory has detected a fault.

6) Technical characteristics

N.B.: all technical specifications refer to a temperature of 20°C.

Electronic control unit

Power supply	: 230Vac (+10-15%) 50Hz
Output maximum power	: 500W / 400VA
Operating temperature	: -20÷55 °C
Dimensions / weight	: 98 x 26 x 20 / 45g
Protection class	: IP54 (container undamaged)
Timer durations	: Factory setting from 1 second to 9 hours TIMER1=1 minute TIMER 2 = 10 minutes

Radio receiver

Frequency	: 433.92 MHz
Coding	: FLO (fixed code), FLOR (rolling code) SMILO (rolling code)
No. of transmitter that can be memorised	: 30
Range of the transmitters	: estimated 150 m outdoor, 20 m indoor. *

* The capacity of the transmitters is strongly influenced by other devices with continuous transmissions which operate at the same frequency. These include alarms, headphones, etc... which interfere with the control unit receiver.

Nice S.p.a reserves the right to make changes to its products at any time as it deems necessary

Declaration of conformity

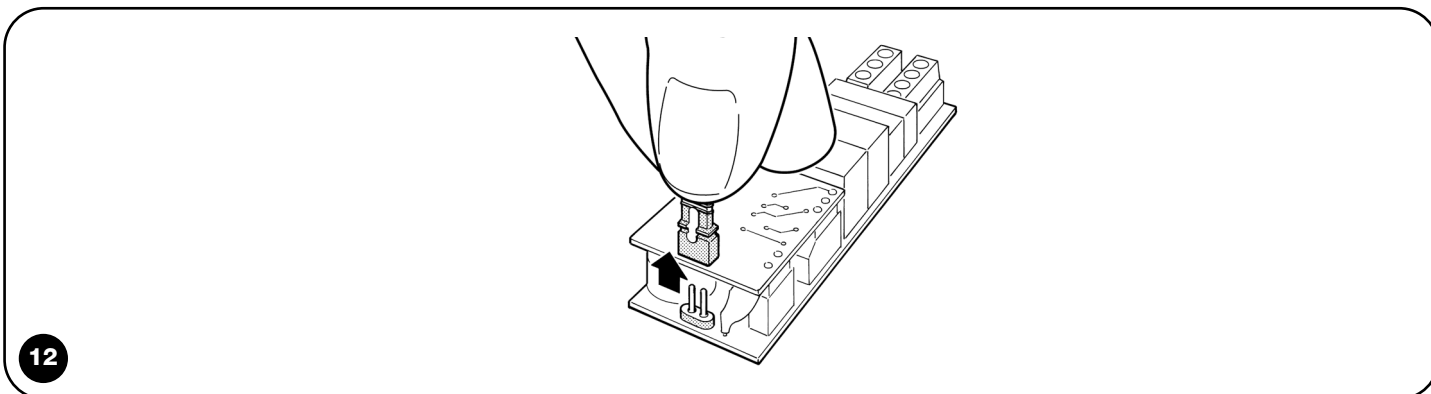
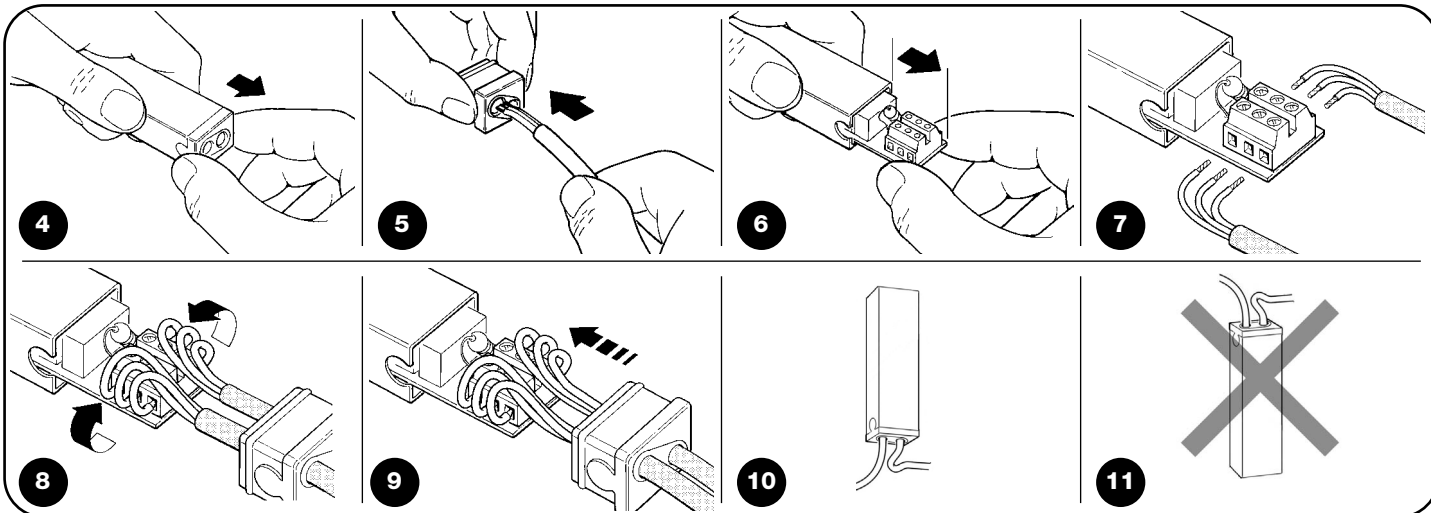
N°: mindy TT1L Rev 0

Nice S.p.a. via Pezza Alta, 13 Rustignè Oderzo (TV) ITALY declares that the product: “mindy TT1L” comply with the essential provisions of European Directive R&TTE 1999/5/EEC, for the uses for which they have been manufactured

Date
22-01-2004

Managing director
Lauro Buoro





COMPANY
WITH QUALITY SYSTEM
CERTIFIED BY DNV
=ISO 9001/2000=

Nice SpA
Oderzo TV Italia
Tel. +39.0422.85.38.38
Fax +39.0422.85.35.85
info@niceforyou.com

Nice Padova
Sarmeola di Rubano PD Italia
Tel. +39.049.89.78.93.2
Fax +39.049.89.73.85.2
info.pd@niceforyou.com

Nice Roma
Roma Italia
Tel. +39.06.72.67.17.61
Fax +39.06.72.67.55.20
info.roma@niceforyou.com

Nice France
Buchelay
Tel. +33.(0)1.30.33.95.95
Fax +33.(0)1.30.33.95.96
info@nicefrance.fr

Nice Rhône-Alpes
Decines Charpieu France
Tel. +33.(0)4.78.26.56.53
Fax +33.(0)4.78.26.57.53
info.lyon@nicefrance.fr

Nice France Sud
Aubagne France
Tel. +33.(0)4.42.62.42.52
Fax +33.(0)4.42.62.42.50
info.marseille@nicefrance.fr

Nice Belgium
Leuven (Heverlee)
Tel. +32.(0)16.38.69.00
Fax +32.(0)16.38.69.01
info@nicebelgium.be

Nice España Madrid
Tel. +34.9.16.16.33.00
Fax +34.9.16.16.30.10

Nice España Barcelona
Tel. +34.9.35.88.34.32
Fax +34.9.35.88.42.49

Nice Polska
Pruszków
Tel. +48.22.728.33.22
Fax +48.22.728.25.10
info@nice.com.pl

Nice China
Shanghai
Tel. +86.21.525.706.34
Fax +86.21.621.929.88
info@niceforyou.com.cn

www.niceforyou.com

Nice Gate is the doors and gate automation division of Nice

Nice Screen is the rolling shutters and awnings automation division of Nice